

Software Requirements Specification

Library Management System

Version 1.0 – Approved
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Revision History

Name	Date	Reason For Changes	Version
Senior Business Analyst	09 June 2026	Initial creation of the SRS for the Library Management System based on the project idea and IEEE-style template	1.0

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and nonfunctional requirements for the Library Management System (LMS), a responsive web-based platform used to manage book catalogs, members, borrowing transactions, returns, overdue processing, late fines, and administrative controls for a library.

This SRS is intended to:

- Establish a shared understanding of the system scope and expected behavior.
- Serve as the baseline for solution design, development, testing, deployment, and acceptance.
- Define detailed user roles, workflows, business rules, interfaces, constraints, and quality expectations.
- Provide traceable requirements for business stakeholders, project managers, designers, developers, testers, and support teams.

This document covers the first production release of the Library Management System, including guest catalog access, librarian operations, administrative controls, fine-rule configuration, reporting, bilingual support, and responsive browser-based access.

1.2 Document Conventions

The following conventions are used throughout this document:

LMS refers to the Library Management System.

Requirements are uniquely identified using the format REQ-[DOMAIN]-[NUMBER].

Business rules are identified using the format BR-[NUMBER].

The terms shall, must, and will indicate mandatory behavior.

The terms may and should indicate optional or recommended behavior.

Role names are capitalized when referring to system-defined roles: Guest, Member, Librarian, and Admin.

1.3 Intended Audience and Reading Suggestions

This document is intended for the following audiences:

Business Sponsors and Product Owners: Review Sections 1, 2, 4, and 5 to validate scope and business outcomes.

Business Analysts: Review the full document for traceability, completeness, and requirement consistency.

Solution Architects and Developers: Focus on Sections 2, 3, 4, and 5 for implementation details and constraints.

UI/UX Designers: Focus on Sections 2.3, 3.1, 4, and 5.4 for user experience, roles, and bilingual requirements.

Quality Assurance and Test Teams: Focus on Sections 4 and 5, especially the functional requirements and acceptance criteria.

System Administrators and Support Teams: Focus on Sections 3, 4.7, and 5 for configuration, security, and operational requirements.

Training and Documentation Teams: Focus on Sections 2.6, 3.1, and 4 for user guidance needs.

Recommended reading order:

1. Section 1 for context and scope.
2. Section 2 for overall product understanding.
3. Section 4 for detailed system behavior.
4. Section 5 for quality and operational expectations.
5. Appendices for definitions, models, and open items.

1.4 Product Scope

The Library Management System is a browser-based application intended to improve library operations by digitizing catalog management, member administration, borrowing, returns, overdue tracking, fine calculation, and reporting.

The product will:

- Enable visitors to browse and search the library catalog without an account.
- Enable registered users to authenticate when borrowing-related access is required.
- Enable Librarians to manage books, members, borrowing, and returns.
- Enable Admins to manage accounts, categories, fine rules, and system settings.
- Provide a dashboard and analytics to support operational monitoring and decision-making.
- Support both Arabic and English interfaces.
- Deliver a responsive user interface that works on desktop and mobile browsers without requiring a native mobile application.

The business objective is to reduce manual recordkeeping, improve book circulation control, provide visibility into availability and overdue items, and ensure consistent fine-rule enforcement.

1.5 References

- Project idea brief for Library Management System, provided by the requester.
- Uploaded SRS template: srs_template-ieee.doc, based on IEEE-style software requirements structure.
- IEEE 830 style of Software Requirements Specification structure, as requested by the requester.
- Standard library circulation practices and common requirements engineering conventions used for business web applications.

2. Overall Description

2.1 Product Perspective

The Library Management System is a new web application intended to replace or improve manual or fragmented library operations. It is envisioned as a centralized platform that manages the core lifecycle of books and borrowing transactions.

At a high level, the system consists of the following logical modules:

- Authentication and role-based access control
- Catalog browsing and search
- Book and inventory management
- Member management
- Borrowing management
- Return and fine calculation engine
- Administrative configuration
- Reporting and analytics
- Bilingual user interface services

The LMS is expected to interact with persistent data storage for books, members, transactions, configurations, and reports. The system is designed as a responsive web application, accessible from supported browsers on desktops, tablets, and smartphones.

2.2 Product Functions

The product shall provide the following major functions:

- Allow Guests to browse and search the catalog.
- Display book details including title, author, category, copy counts, and availability.
- Prompt unauthenticated users to log in when attempting restricted actions.
- Allow Librarians to create, update, and deactivate book records.
- Allow Librarians to manage member profiles and membership details.
- Record borrowing transactions with member, book, issue date, and expected return date.
- Display all active borrowings and borrowing history.
- Record returns and automatically calculate late fines based on configurable rules.
- Update book availability in real time when items are borrowed or returned.
- Allow Admins to manage user accounts, librarian roles, categories, and system settings.
- Allow Admins to configure daily fine values and related policy parameters.
- Provide dashboards and reports for borrowed books, overdue items, fines, stock availability, and member activity.
- Support Arabic and English languages.
- Enforce role-based access restrictions across all secured functions.

2.3 User Classes and Characteristics

Guest

- Unauthenticated public user.
- Can browse and search the catalog and view book details and availability.
- Cannot borrow books or access restricted operations.
- Prompted to log in or register when a restricted action is attempted.

Member

- Registered library patron with an active membership profile.
- Can authenticate to view personal borrowing information, active loans, due dates, and outstanding fines.
- Does not manage books, categories, or platform settings.
- Has basic digital literacy and may access the system via desktop or mobile browser.

Librarian

- Operational library staff responsible for day-to-day circulation tasks.
- Can manage books, member records, borrowing transactions, and returns.
- Needs quick workflows, validation prompts, and error prevention controls.
- Requires moderate system training.

Admin

- System or library administrator with elevated permissions.
- Can manage librarian accounts, role assignments, categories, fine rules, and system settings.
- Can access operational and management reports.
- Requires controlled access to sensitive configurations and audit-visible changes.

2.4 Operating Environment

Client Devices: Desktop computers, laptops, tablets, and smartphones.

Client Browsers: Current and previous major versions of Chrome, Edge, Firefox, and Safari.

Application Type: Responsive web application.

Server Environment: Web application server hosted on cloud or on-premises infrastructure.

Database: Relational database management system or equivalent transactional data store.

Network: Internal network and/or public internet over secure HTTPS.

Language Support: Arabic and English.

Timezone Handling: Configurable library-local timezone.

2.5 Design and Implementation Constraints

- The solution shall be delivered as a web-based platform; no native mobile application is in scope for Release 1.0.
- The user interface shall support Arabic and English with proper directional rendering for LTR and RTL layouts.
- Access control shall be role-based.
- Fine calculation shall be configurable and consistently applied using system rules.

- The system shall maintain reliable transactional integrity for borrowing and return operations.
- Responsive design shall be supported for mobile browser usage.
- The initial release is assumed to support a single library organization unless expanded in a later phase.
- Integration with payment gateways, barcode scanners, RFID devices, or national identity services is not mandatory for Release 1.0.

2.6 User Documentation

- User quick-start guide for Guests and Members.
- Librarian operations manual covering book, member, borrowing, and return workflows.
- Admin guide covering user administration, fine rule configuration, category management, and reports.
- On-screen validation messages and contextual help for major forms.
- FAQ and support guidance for login, search, borrowing status, and overdue handling.
- Training materials for Arabic and English users, where applicable.

2.7 Assumptions and Dependencies

1. The library operates with one centralized catalog in Release 1.0.
 1. A Member role is assumed even though the brief explicitly lists Guest, Librarian, and Admin, because users are required to log in for borrowing-related access.
 1. Borrowing and return transactions are operationally processed by Librarians, while Members may view their own status online.
 1. Fine calculation is based on a configurable amount per overdue day and begins after the expected return date has passed.
 1. Fine payment processing is out of scope; the system will track outstanding fines and settlement status but not integrate with a payment gateway in Release 1.0.
 1. Each book title may have multiple copies. Availability is tracked using total copies and currently borrowed copies; copy-level barcode control may be added later.
 1. Notification channels such as email or SMS are not mandatory for Release 1.0.
- Historical migration from legacy systems is not in scope unless separately approved.

3. External Interface Requirements

3.1 User Interfaces

The system shall provide the following user interfaces:

Public / Guest Interfaces

- Home page with catalog highlights and search bar.
- Catalog browsing page with filters and availability indicators.
- Book details page with title, author, category, description, and current availability.
- Login prompt when restricted actions are attempted.

Member Interfaces

- Secure login page.
- Personal dashboard showing active loans, due dates, overdue items, and outstanding fines.
- Borrowing history page.
- Profile view page.

Librarian Interfaces

- Dashboard with circulation summary.
- Book management screens for add, edit, view, archive, and search.
- Member management screens for registration, update, search, and history review.
- Borrowing transaction form with validation and due-date assignment.
- Return processing screen with auto-calculated late fine display.
- Active borrowings and overdue list screens.

Admin Interfaces

- User management screen for librarian/admin accounts and role assignments.
- Fine rule configuration screen.
- Category management screen.
- Reports and analytics dashboard.
- Platform settings screen.

General UI Requirements

- The UI shall be responsive across desktop and mobile browsers.
- The UI shall support Arabic and English language switching.
- Labels, validations, and messages shall be localized.
- Availability status shall be visually clear.
- Sensitive actions shall require confirmation or appropriate privilege checks.
- Lists shall support sorting, filtering, pagination, and clear empty-state messaging.
- Forms shall provide mandatory-field indicators and user-friendly validation errors.

3.2 Hardware Interfaces

No specialized hardware interface is required for Release 1.0. The system shall support standard hardware environments including desktop and laptop computers, tablets and smartphones, and standard printers for optional report printing. Potential future hardware integrations (not in current scope) may include barcode scanners, RFID readers, receipt printers, and self-service kiosks.

3.3 Software Interfaces

The system shall interface with the following software components:

- Database layer for storing books, members, users, transactions, settings, and audit records.
- Authentication service for login, password management, and session handling.
- Localization service to render Arabic and English text.
- Reporting engine for dashboard aggregation and report generation.
- Logging/audit service for security and operational traceability.

3.4 Communications Interfaces

- The system shall use HTTPS for all web-based communication in production.
- Client-server communication shall support secure session management.
- The application shall support standard web protocols over TCP/IP.
- Timeout, retry, and error-handling behavior shall be implemented for critical data submission operations.

4. System Features

4.1 Authentication and Role-Based Access Control

4.1.1 Description and Priority

This feature controls secure access to the system and restricts functionality based on user role. It is foundational to all secured operations and ensures that Guests, Members, Librarians, and Admins only access functions appropriate to their privileges.

Priority: High

4.1.2 Stimulus/Response Sequences

- 1. Stimulus:** A Guest selects a restricted action such as borrow-related access. **Response:** The system prompts the user to log in and redirects to the authentication page.
- 2. Stimulus:** A valid user submits correct login credentials. **Response:** The system authenticates the user, starts a secure session, and displays the role-appropriate landing page.
- 3. Stimulus:** A user submits invalid credentials. **Response:** The system denies access, displays a generic error message, and logs the attempt.
- 4. Stimulus:** A user attempts to access a function outside their role privileges. **Response:** The system blocks access, displays an authorization error, and records the event.
- 5. Stimulus:** A user remains inactive beyond the configured timeout period. **Response:** The system terminates the session and requires re-authentication.

4.1.3 Functional Requirements

REQ-AUTH-001: The system shall support secure login for Members, Librarians, and Admins using unique credentials.

Acceptance Criteria: A registered active user with valid credentials can sign in and reach the correct dashboard.

REQ-AUTH-002: The system shall prevent unauthenticated users from accessing restricted pages, APIs, and transactions.

Acceptance Criteria: Attempted direct URL access by an unauthenticated user results in redirect to login or access denial.

REQ-AUTH-003: The system shall enforce role-based access control so that only authorized roles can perform privileged actions.

Acceptance Criteria: Guests cannot manage data; Members cannot administer users; Librarians cannot alter restricted system settings; Admins can access configured administration functions.

REQ-AUTH-004: The system shall display role-appropriate navigation menus and landing pages after login.

Acceptance Criteria: Each role sees only relevant menu items and default pages.

REQ-AUTH-005: The system shall support logout and session termination.

Acceptance Criteria: After logout, secured pages are inaccessible until the user logs in again.

REQ-AUTH-006: The system shall expire inactive sessions after a configurable timeout period.

Acceptance Criteria: After the timeout threshold is reached, the session is invalidated and the user is prompted to log in again.

REQ-AUTH-007: The system shall log authentication events and authorization failures for audit purposes.

Acceptance Criteria: Successful login, failed login, logout, and access-denied events are traceable in audit logs.

REQ-AUTH-008: The system shall allow Admins to activate, deactivate, or reset access for Librarian and Admin accounts.

Acceptance Criteria: Deactivated users cannot log in, and reset actions are logged.

4.2 Catalog Browsing and Search

4.2.1 Description and Priority

This feature enables Guests, Members, Librarians, and Admins to browse and search the library catalog efficiently. It supports title discovery, availability visibility, and filtering by common search criteria.

Priority: High

4.2.2 Stimulus/Response Sequences

- 1. Stimulus:** A Guest opens the catalog page. **Response:** The system displays a browsable list of books and available filters.
- 2. Stimulus:** A user enters a keyword or applies filters. **Response:** The system returns matching books based on title, author, category, and availability criteria.
- 3. Stimulus:** A user selects a book record. **Response:** The system displays the book detail page with descriptive and availability information.
- 4. Stimulus:** No books match the criteria. **Response:** The system displays an empty-state message and allows criteria refinement.

4.2.3 Functional Requirements

REQ-SEARCH-001: The system shall allow catalog browsing without login.

Acceptance Criteria: Guests can view catalog pages and book details without authentication.

REQ-SEARCH-002: The system shall support keyword search by book title, author, and category.

Acceptance Criteria: Search results include relevant matches across the configured searchable fields.

REQ-SEARCH-003: The system shall allow users to filter books by availability status.

Acceptance Criteria: Users can refine results to available, borrowed, or all books.

REQ-SEARCH-004: The system shall display book details including title, author, category, summary, total copies, and available copies.

Acceptance Criteria: The detail page displays the latest stored values for the selected book.

REQ-SEARCH-005: The system shall support pagination or lazy loading for large result sets.

Acceptance Criteria: Result lists remain usable and performant when the catalog contains many records.

REQ-SEARCH-006: The system shall preserve applied filters and search criteria during the user's current navigation session where technically appropriate.

Acceptance Criteria: Returning to the result list restores the recent search context.

REQ-SEARCH-007: The system shall present search and catalog information in the selected interface language.

Acceptance Criteria: Labels, messages, and static UI text change correctly between Arabic and English.

4.3 Book Catalog and Inventory Management

4.3.1 Description and Priority

This feature enables Librarians to maintain the library book catalog and copy counts, and ensures the system accurately distinguishes available and borrowed quantities.

Priority: High

4.3.2 Stimulus/Response Sequences

- 1. Stimulus:** A Librarian selects "Add Book". **Response:** The system displays a book entry form and saves the record after successful validation.
- 2. Stimulus:** A Librarian updates a book record. **Response:** The system validates the changes, saves the updated values, and displays confirmation.
- 3. Stimulus:** A Librarian attempts to delete or deactivate a book that is currently borrowed. **Response:** The system prevents invalid deletion and requires a safe status handling approach.
- 4. Stimulus:** A borrow or return transaction occurs. **Response:** The system automatically adjusts the available-versus-borrowed copy counts.

4.3.3 Functional Requirements

REQ-BOOK-001: The system shall allow authorized Librarians to create book records with at least title, author, category, and number of copies.

Acceptance Criteria: Required fields are validated and a new book record is persisted.

REQ-BOOK-002: The system shall allow authorized Librarians to edit book metadata and copy counts.

Acceptance Criteria: Updated values appear in catalog views and downstream operations.

REQ-BOOK-003: The system shall allow authorized Librarians to deactivate or archive books no longer available for circulation.

Acceptance Criteria: Archived books are excluded from active circulation while historical transactions remain intact.

REQ-BOOK-004: The system shall prevent the available copy count from becoming negative.

Acceptance Criteria: Borrowing is blocked if no available copies remain.

REQ-BOOK-005: The system shall calculate and display total copies, borrowed copies, and available copies for each book.

Acceptance Criteria: Inventory counts update immediately following successful borrow and return transactions.

REQ-BOOK-006: The system shall link each book to a valid category managed by the system.

Acceptance Criteria: Book creation or update cannot be completed with an invalid or inactive category.

REQ-BOOK-007: The system shall retain historical transaction references even if a book is later archived.

Acceptance Criteria: Past borrowing records remain visible and traceable.

REQ-BOOK-008: The system shall validate duplicate creation risk based on configurable identifying attributes such as title, author, or ISBN.

Acceptance Criteria: The system warns users when a likely duplicate record is being created.

4.4 Member Management

4.4.1 Description and Priority

This feature enables Librarians to register and maintain member profiles and allows authorized users to view borrowing history, active borrows, and outstanding fines.

Priority: High

4.4.2 Stimulus/Response Sequences

- 1. Stimulus:** A Librarian selects "Register Member". **Response:** The system displays a member form and creates the profile after validation.
- 2. Stimulus:** A Librarian searches for a member. **Response:** The system displays matching member records and key status indicators.
- 3. Stimulus:** A Librarian opens a member profile. **Response:** The system displays profile details, active borrowings, borrowing history, and outstanding fines.
- 4. Stimulus:** A member profile is suspended or deactivated. **Response:** The system blocks new borrow transactions while preserving history.

4.4.3 Functional Requirements

REQ-MEMBER-001: The system shall allow authorized Librarians to create member profiles with at least name, contact information, and membership details.

Acceptance Criteria: A member profile cannot be saved without mandatory fields.

REQ-MEMBER-002: The system shall allow authorized users to update member contact and membership information.

Acceptance Criteria: Updates are stored with audit visibility where applicable.

REQ-MEMBER-003: The system shall support member search using key fields such as name, membership identifier, phone number, or email.

Acceptance Criteria: Search results return relevant matches within acceptable response time.

REQ-MEMBER-004: The system shall display each member's active borrows, full borrowing history, overdue items, and outstanding fines.

Acceptance Criteria: The member profile screen provides a consolidated circulation view.

REQ-MEMBER-005: The system shall allow authorized users to suspend or deactivate a member record.

Acceptance Criteria: Suspended or inactive members cannot initiate new borrow transactions.

REQ-MEMBER-006: The system shall maintain a unique identifier for each member profile.

Acceptance Criteria: Duplicate active member identities are controlled according to configured validation rules.

REQ-MEMBER-007: The system shall allow an authenticated Member to view their own profile-level borrowing and fine information without exposing other members' data.

Acceptance Criteria: A Member only sees their own data and receives access denial when attempting to access another member's record.

4.5 Borrowing Management

4.5.1 Description and Priority

This feature enables Librarians to record borrowing transactions, assign expected return dates, and ensure that only eligible members can borrow available books.

Priority: High

4.5.2 Stimulus/Response Sequences

- 1. Stimulus:** A Librarian selects a member and a book for borrowing. **Response:** The system validates eligibility, availability, and borrowing rules.
- 2. Stimulus:** The Librarian confirms the transaction and enters or accepts a due date. **Response:** The system creates the borrowing record, reduces available copies, and confirms success.
- 3. Stimulus:** The selected member is inactive or has unresolved borrowing restrictions. **Response:** The system blocks the transaction and displays the reason.
- 4. Stimulus:** No copies are available. **Response:** The system prevents the borrowing transaction.

4.5.3 Functional Requirements

REQ-BORROW-001: The system shall allow authorized Librarians to record a borrowing transaction linked to a valid member and a valid book.

Acceptance Criteria: A transaction record is created only when both references are valid and active.

REQ-BORROW-002: The system shall require an expected return date for every borrowing transaction.

Acceptance Criteria: The transaction cannot be completed without a due date.

REQ-BORROW-003: The system shall verify that at least one copy is available before a borrow transaction is committed.

Acceptance Criteria: Borrowing is blocked when available copies equal zero.

REQ-BORROW-004: The system shall verify that the member is eligible to borrow based on active status and configured library rules.

Acceptance Criteria: Ineligible members receive a clear blocking reason.

REQ-BORROW-005: The system shall decrement available copy count and increment borrowed count after a successful borrow transaction.

Acceptance Criteria: Inventory values change immediately after the transaction is saved.

REQ-BORROW-006: The system shall record borrow transaction details including transaction identifier, member, book, issue date, expected return date, status, and processing user.

Acceptance Criteria: The stored record supports operational review and audit traceability.

REQ-BORROW-007: The system shall allow authorized users to view all currently borrowed books and filter them by member, due date, or overdue status.

Acceptance Criteria: Operational users can retrieve active loan records quickly.

REQ-BORROW-008: The system shall prevent duplicate active borrowing of the same copy or same title beyond allowed availability rules.

Acceptance Criteria: The system enforces inventory consistency and prevents invalid duplicate active loans.

4.6 Return Processing and Fine Calculation

4.6.1 Description and Priority

This feature enables Librarians to record returns, restore book availability, and automatically calculate overdue fines according to configurable rules.

Priority: High

4.6.2 Stimulus/Response Sequences

- 1. Stimulus:** A Librarian selects an active borrowing record for return. **Response:** The system retrieves the transaction details and calculates whether the item is overdue.
- 2. Stimulus:** The return date is recorded. **Response:** The system updates the loan status to returned, restores book availability, and calculates any fine due.
- 3. Stimulus:** The book is returned on or before the due date. **Response:** The system records the return with zero overdue fine.
- 4. Stimulus:** The book is returned after the due date. **Response:** The system calculates overdue days and the corresponding fine based on configured fine rules.

4.6.3 Functional Requirements

REQ-RETURN-001: The system shall allow authorized Librarians to record the return of an active borrowing transaction.

Acceptance Criteria: Only active loans can be processed as returns.

REQ-RETURN-002: The system shall calculate overdue status by comparing the actual return date against the expected return date.

Acceptance Criteria: The system consistently identifies overdue versus on-time returns.

REQ-RETURN-003: The system shall automatically calculate late fines using the configured daily fine amount and the number of overdue days.

Acceptance Criteria: The fine amount equals overdue days multiplied by the configured per-day fine value.

REQ-RETURN-004: The system shall update the book's inventory counts immediately after a successful return.

Acceptance Criteria: Borrowed count decreases and available count increases accordingly.

REQ-RETURN-005: The system shall store the calculated fine amount and outstanding status in the member account history.

Acceptance Criteria: Outstanding fine balances are visible on the member profile and reports.

REQ-RETURN-006: The system shall support zero-fine processing for on-time returns.

Acceptance Criteria: Returned-on-time items do not generate overdue charges.

REQ-RETURN-007: The system shall maintain an immutable historical record of the original due date, actual return date, calculated overdue days, and applied fine.

Acceptance Criteria: Historical return records remain available for audit and reporting.

REQ-RETURN-008: The system shall allow authorized handling of fine settlement status and administrative adjustments where permitted by policy.

Acceptance Criteria: Only authorized roles can mark fines as settled or adjusted, and all such actions are auditable.

4.7 Administration and Configuration

4.7.1 Description and Priority

This feature enables Admins to manage platform users, librarian roles, fine rules, categories, and key operational settings.

Priority: High

4.7.2 Stimulus/Response Sequences

- 1. Stimulus:** An Admin opens user administration. **Response:** The system displays current user accounts and role assignments.
- 2. Stimulus:** An Admin creates or updates a librarian account. **Response:** The system validates the action, saves the account, and records the change.
- 3. Stimulus:** An Admin changes the fine amount per day. **Response:** The system saves the new rule and uses it for future overdue calculations.
- 4. Stimulus:** An Admin manages categories. **Response:** The system creates, updates, activates, or deactivates categories subject to validation.

4.7.3 Functional Requirements

REQ-ADMIN-001: The system shall allow Admins to create, update, activate, deactivate, and assign roles for librarian and administrative user accounts.

Acceptance Criteria: Role-managed users can only access capabilities assigned to their roles.

REQ-ADMIN-002: The system shall allow Admins to configure fine rules, including at minimum the fine amount per overdue day.

Acceptance Criteria: New overdue calculations use the latest approved active configuration.

REQ-ADMIN-003: The system shall allow Admins to manage book categories.

Acceptance Criteria: Categories can be created, edited, activated, or deactivated without corrupting linked historical records.

REQ-ADMIN-004: The system shall allow Admins to configure core platform settings such as default language, session timeout, and operational parameters.

Acceptance Criteria: Changes are persisted and applied according to configuration scope.

REQ-ADMIN-005: The system shall log all administrative configuration changes with user, timestamp, and before/after values where feasible.

Acceptance Criteria: Audit records exist for critical configuration changes.

REQ-ADMIN-006: The system shall prevent unauthorized roles from accessing administrative configuration functions.

Acceptance Criteria: Librarians, Members, and Guests cannot use Admin-only screens unless explicitly granted.

REQ-ADMIN-007: The system shall validate configuration data to prevent invalid operational settings.

Acceptance Criteria: Negative fine values, duplicate active categories, or invalid timeout settings are rejected.

4.8 Dashboard, Reporting, and Analytics

4.8.1 Description and Priority

This feature provides operational visibility and management insight through dashboards and reports related to circulation, inventory, overdue items, fines, and member activity.

Priority: Medium-High

4.8.2 Stimulus/Response Sequences

- 1. Stimulus:** A Librarian or Admin opens the dashboard. **Response:** The system displays role-appropriate summary metrics and recent operational information.
- 2. Stimulus:** A user selects a specific report. **Response:** The system retrieves and displays the requested report using current or filtered data.
- 3. Stimulus:** A user filters by date range, category, or status. **Response:** The system recalculates the report output and displays updated results.
- 4. Stimulus:** A user exports a report where export is enabled. **Response:** The system generates the report in the configured downloadable format.

4.8.3 Functional Requirements

REQ-REPORT-001: The system shall provide dashboard widgets showing currently borrowed books, overdue books, outstanding fines, and total available books.

Acceptance Criteria: Dashboard values match the underlying transactional data at the time of retrieval.

REQ-REPORT-002: The system shall provide a report of currently borrowed books.

Acceptance Criteria: The report lists active loans with member, book, due date, and status information.

REQ-REPORT-003: The system shall provide an overdue books and outstanding fines report.

Acceptance Criteria: Users can identify overdue items, overdue days, and associated fine amounts.

REQ-REPORT-004: The system shall provide a most-borrowed-books report over a selectable period.

Acceptance Criteria: Books are ranked by borrow frequency using stored transaction history.

REQ-REPORT-005: The system shall provide a member activity report showing borrowing trends and member-level circulation history.

Acceptance Criteria: The report supports operational review of member usage patterns.

REQ-REPORT-006: The system shall restrict report visibility based on role and data sensitivity.

Acceptance Criteria: Users only see reports authorized for their role.

REQ-REPORT-007: The system shall support filtering of reports by relevant dimensions such as date range, category, status, or member.

Acceptance Criteria: Report outputs refresh correctly based on applied filters.

REQ-REPORT-008: The system shall support report export in at least one standard downloadable format in production configuration, if enabled by the organization.

Acceptance Criteria: Exported reports match displayed content and are downloadable successfully.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

NFR-PERF-001: The system shall return standard catalog search results within 3 seconds for normal expected loads.

NFR-PERF-002: The system shall complete standard create, update, borrow, and return transactions within 3 seconds under normal operating conditions.

NFR-PERF-003: Dashboard summary data shall load within 5 seconds under normal operating conditions.

NFR-PERF-004: The system shall support concurrent users appropriate to the organization's defined capacity without data corruption or unacceptable degradation.

NFR-PERF-005: Time-sensitive inventory and transaction updates shall be committed atomically to prevent inconsistent availability counts.

NFR-PERF-006: The application shall use pagination, indexing, caching, or equivalent techniques where needed to maintain acceptable performance.

5.2 Safety Requirements

Although the system is not a life-critical application, the following safety-related operational safeguards apply:

NFR-SAFE-001: The system shall prevent accidental data loss through confirmation prompts for destructive administrative actions where appropriate.

NFR-SAFE-002: The system shall preserve historical transaction records even when book, category, or member records are deactivated.

NFR-SAFE-003: The system shall validate critical transaction steps to prevent erroneous borrowing or return posting.

NFR-SAFE-004: The system shall provide recoverable auditability for high-impact operational changes.

5.3 Security Requirements

NFR-SEC-001: The system shall require authentication for all restricted functions.

NFR-SEC-002: The system shall enforce role-based authorization on both user interface and backend service levels.

NFR-SEC-003: The system shall protect credentials and sessions using accepted secure practices.

NFR-SEC-004: The system shall use HTTPS in production.

NFR-SEC-005: The system shall protect personal and borrowing-related data from unauthorized disclosure.

NFR-SEC-006: The system shall log security-relevant events such as failed logins, access denials, and critical configuration changes.

NFR-SEC-007: The system shall support password reset and account recovery processes under controlled administrative or self-service policy.

NFR-SEC-008: The system shall validate user input to reduce risks such as injection attacks, invalid data submission, and unauthorized manipulation.

NFR-SEC-009: The system shall enforce session timeout and secure logout.

NFR-SEC-010: The system shall ensure that Members cannot access other members' personal or circulation data.

5.4 Software Quality Attributes

Usability

- The system shall provide intuitive navigation for Guests, Members, Librarians, and Admins.
- Form validations and error messages shall be clear and actionable.
- Common librarian workflows shall minimize unnecessary clicks.

Reliability

- The system shall maintain data consistency for inventory and transaction records.
- Failed transactions shall not leave the system in a partially committed state.

Availability

- The system shall be designed for business-hour availability at minimum, with deployment targets defined by the hosting organization.

Maintainability

- The solution shall support modular enhancement of catalog, circulation, reporting, and administration features.
- Configuration-driven fine rules and category management shall reduce the need for code changes.

Scalability

- The solution shall support growth in catalog size, member volume, and transaction history.

Accessibility

- The UI shall use readable text, clear contrast, and structured layouts appropriate for common accessibility practices.
- The responsive design shall remain usable on smaller screens.

Localization

- The system shall support both Arabic and English labels, messages, and navigation behavior.
- Right-to-left rendering shall be correctly applied for Arabic screens.

5.5 Business Rules

BR-001: Only authenticated and eligible users may access restricted borrowing-related functions.

BR-002: Guests may browse and search the catalog but may not borrow books.

BR-003: A book cannot be borrowed if no available copies remain.

BR-004: Every borrowing transaction must have an expected return date.

BR-005: Late fines are calculated based on the configured amount per overdue day.

BR-006: Fine rules are controlled by Admins and applied consistently to future relevant transactions.

BR-007: Member profiles must exist and be active before new borrow transactions can be recorded.

BR-008: Returned books immediately restore availability counts.

BR-009: Historical borrowing and return records must remain available for audit and reporting.

BR-010: Only authorized roles may alter categories, fine rules, system settings, or privileged user accounts.

BR-011: Suspended or inactive members cannot initiate new borrow transactions.

BR-012: The system must support Arabic and English as official interface languages.

BR-013: Destructive or high-impact administrative actions shall be auditable.

BR-014: Outstanding fines remain associated with the member record until settled or adjusted by authorized process.

5.6 Other Requirements

- The solution shall present a consistent visual experience across supported browsers and devices.
- Date, currency, and language presentation shall follow configurable locale settings where applicable.
- The system shall be deployable in environments suitable for library operations with standard backup and recovery procedures.
- The system shall support future extensibility for features such as notifications, barcode integration, payment processing, and multi-branch support.
- Data export, if enabled, shall respect authorization and privacy constraints.

Appendix A: Glossary

Term	Definition
Admin	Privileged user responsible for system configuration, user administration, categories, and fine-rule management.
Availability	The number or status of copies currently available for borrowing.
Book Category	Classification assigned to a book, such as Fiction, Science, or History.
Borrowing Transaction	A recorded event where a book is issued to a member with an expected return date.
Catalog	The searchable collection of books maintained by the system.
Fine	Monetary penalty calculated for overdue returns according to configured rules.
Guest	Unauthenticated user who may browse and search the catalog only.
Librarian	Operational user responsible for books, members, borrowing, and returns.
LMS	Library Management System.
Member	Registered library patron whose profile, borrowing activity, and fines are managed by the system.
Overdue	Status indicating that the actual return date is later than the expected return date.
RBAC	Role-Based Access Control, a security model that restricts functions by user role.
Return Transaction	The process of recording that a previously borrowed book has been returned.
Responsive UI	User interface that adapts to different screen sizes and devices.
SRS	Software Requirements Specification.

Appendix B: Analysis Models

B.1 Use Case Model Summary

Use Case	Primary Actor	Brief Description
Browse Catalog	Guest / Member / Librarian / Admin	Search and browse books by title, author, category, and availability.
View Book Details	Guest / Member / Librarian / Admin	Open a detailed view of a selected book.
Log In	Member / Librarian / Admin	Authenticate and access role-specific functions.
Manage Books	Librarian	Add, edit, archive, and review books and inventory.
Manage Members	Librarian	Register, update, search, suspend, and review member records.
Record Borrowing	Librarian	Issue a book to an eligible member and set expected return date.
Record Return	Librarian	Process a returned book and calculate fine if overdue.
Manage Categories	Admin	Create, update, activate, or deactivate categories.
Manage Users and Roles	Admin	Manage librarian/admin accounts and role permissions.
Configure Fine Rules	Admin	Define fine amount per day and related configuration.
View Reports	Librarian / Admin	Review operational and analytical reports.
View Personal Borrowing Status	Member	Review own loans, due dates, history, and outstanding fines.

B.2 Conceptual Data Model Summary

Entity	Key Attributes	Relationships
User Account	User ID, username, role, status, last login	Linked to authentication and audit events
Member	Member ID, name, contact, membership status	Has many borrowing transactions and fines
Book	Book ID, title, author, category, total copies, available copies, status	Belongs to a category; has many borrowing transactions

Entity	Key Attributes	Relationships
Category	Category ID, name, status	Linked to many books
Borrow Transaction	Transaction ID, member, book, issue date, due date, status, processed by	Linked to one member and one book
Return Detail	Return date, overdue days, fine amount, settlement status	Linked to one borrow transaction
Fine Rule	Rule ID, amount per day, effective date, status	Used by overdue calculation engine
Audit Log	Event ID, user, timestamp, action, before/after details	Linked to security and administration events

B.3 Process Flow Summaries

Borrowing Workflow

2. Librarian searches and selects member.
2. System validates member status and borrowing eligibility.
2. Librarian searches and selects book.
2. System validates book availability.
2. Librarian confirms due date.
2. System creates borrowing record and updates inventory.
2. System displays success confirmation.

Return Workflow

3. Librarian locates active borrowing transaction.
3. System retrieves due date and current status.
3. Librarian confirms return.
3. System records actual return date.
3. System calculates overdue days and fine, if any.
3. System updates inventory and member fine balance.
3. System displays return summary.

Fine Rule Configuration Workflow

4. Admin opens fine settings.
4. Admin enters or updates daily fine amount.
4. System validates configuration.
4. System saves rule and audit entry.
4. System applies rule to subsequent relevant overdue calculations.

B.4 State Model Summary

Book Circulation State

- Available – book can be borrowed.

- Borrowed – book is currently on loan and unavailable for new borrowing.
- Returned – borrowing is closed and the book becomes available again.
- Archived/Inactive – book is not available for new circulation but history remains.

Member Status State

- Active – member can borrow subject to policy.
- Suspended – member cannot initiate new borrowing.
- Inactive – member record retained for history but unavailable for new borrowing.

Appendix C: To Be Determined List

The following items require resolution before or during detailed design and development:

6. Maximum number of books a member may borrow at one time.
7. Default borrowing period by category or member type.
8. Whether different categories require different fine values or loan durations.
9. Whether partial fine waiver is allowed for Librarians or restricted to Admins only.
10. Whether email or SMS reminders for due dates and overdue notices are required in a later release.
11. Whether online fine payment integration is needed.
12. Whether barcode or RFID support is required in a future phase.
13. Whether the system must support multiple branches or separate inventories.
14. Whether ISBN, publisher, edition, and cover image are mandatory or optional metadata fields.
15. Whether report export formats must include PDF, Excel, CSV, or all of them.
16. Whether member self-registration is allowed or all member creation must be librarian-driven.
17. Whether soft delete, archive, or permanent delete behavior is preferred for each master-data object.